Real-time Pipeline & Streaming Application

Background

You have been asked to create a real-time pipeline to ingest/receive data from a real-time stream of events and deploy a streaming application to aggregate some of these events in real-time.

Pageviews

The source of data consists of a real-time feed originating from an operational system which tracks visits to a website. A pageview event has 4 properties embedded in a JSON:

- A `user_id` (integer), uniquely identifying a user. This matches the `id` on the users stream. Example: `1234`
- A `postcode` (string), indicating the user's current location when visiting the website. Example: `SW19`.
- A `webpage` (string), indicating the URL of the page being visited. Example: `www.website.com/index.html`
- A epoch `timestamp` in seconds (bigint) of when the pageview event occurred

A sample event would thus be:

```
    {'user_id': 1234, 'postcode': 'SW19', 'webpage':
'www.website.com/index.html', 'timestamp': 1611662684}
```

A page view event is in itself immutable. Pageview events should be received through a Kinesis stream or Kafka topic and delivered into an S3 bucket or local file system folder in real-time (without any transformations).

For this exercise, we assume that the website is visited less than 100K times per day.

Streaming Application

In addition, we also want to deploy a streaming application which will, in real-time, aggregate pageviews data and deliver these aggregates into a separate subfolder/subpath in your S3 bucket or local filesystem folder. The aggregated data should consist of:

- Postcode
- Datetime of the aggregation window, bucketed into 1m intervals
- Number of pageviews occurred during each time window and postcode

Requirements

General Guidelines

- Include code to produce sample data into the source stream/topic
- Source code should be provided via a publicly accessible Github repository
- Provide basic documentation to run the pipeline, along with any other documentation you think is appropriate
- Be prepared to explain your choices.

AWS/Kinesis implementation

- Make use of localstack or free tier AWS services like Kinesis, Kinesis Data Analytics, S3, etc, anything you deem necessary
- Write all infrastructure as code, preferably using terraform

Kafka

- Make use of any stream processing tech you prefer: Flink, kSQL, etc
- Develop the solution using Docker containers which can be easily executed/tested on a local machine

Extra Mile Bonus (**not** a requirement)

In addition to the above, time permitting, consider the following suggestions for taking your implementation a step further.

- Monitoring/Alerting
- Security
- Automation
- Diagrams